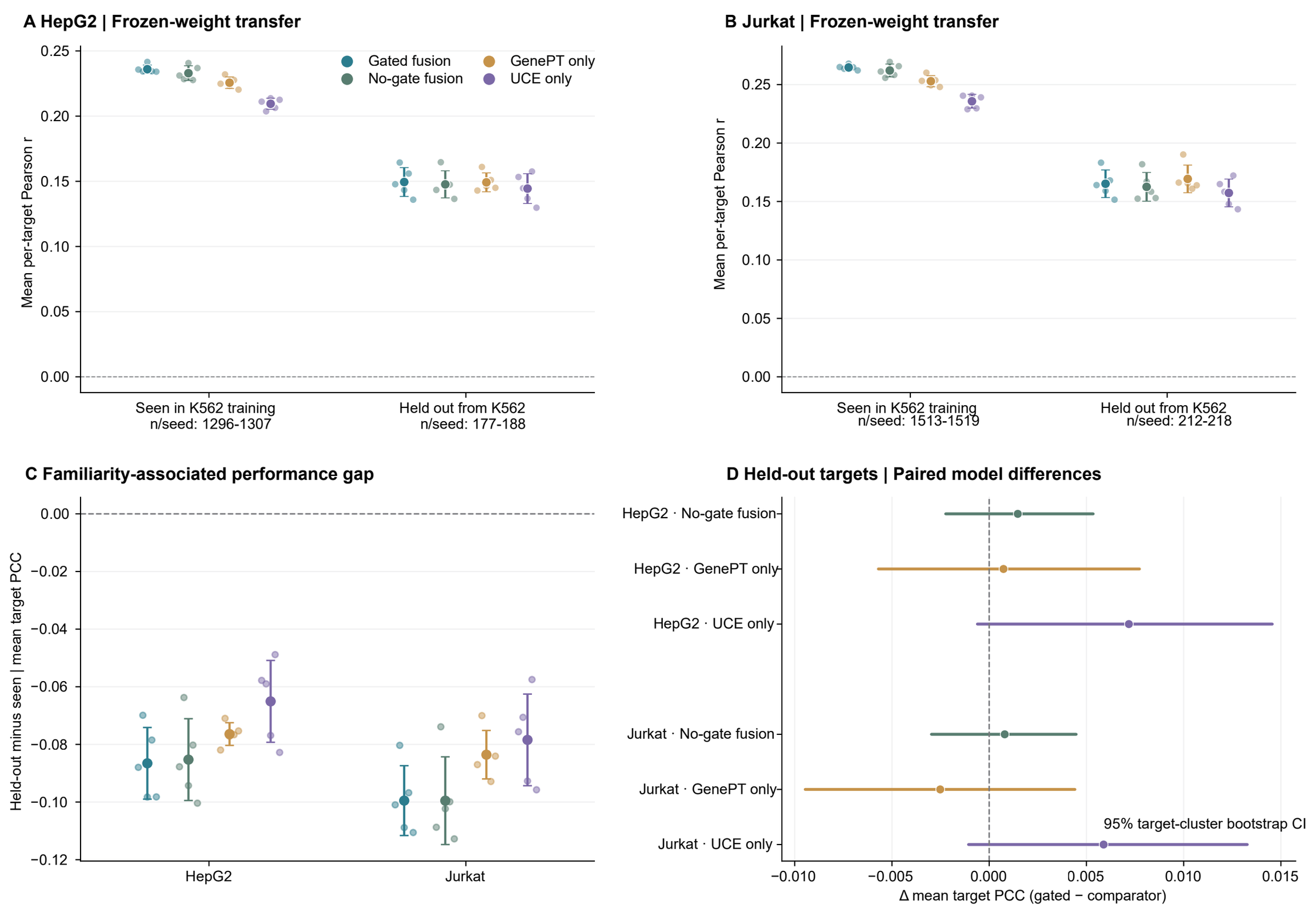

A HepG2 | Frozen-weight transfer
Gated fusion
No-gate fusion
GenePT only
UCE only
Mean per-target Pearson r
0.25
0.20
0.15
0.10
0.05
0.00
Seen in K562 training
n/seed: 1296-1307
Held out from K562
n/seed: 177-188
B Jurkat | Frozen-weight transfer
Mean per-target Pearson r
0.25
0.20
0.15
0.10
0.05
0.00
Seen in K562 training
n/seed: 1513-1519
Held out from K562
n/seed: 212-218
C Familiarity-associated performance gap
Held-out minus seen | mean target PCC
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
−0.12
HepG2
Jurkat
D Held-out targets | Paired model differences
HepG2 · No-gate fusion
HepG2 · GenePT only
HepG2 · UCE only
Jurkat · No-gate fusion
Jurkat · GenePT only
Jurkat · UCE only
95% target-cluster bootstrap CI
−0.010
−0.005
0.000
0.005
0.010
0.015
Δ mean target PCC (gated − comparator)